

times more flexible than any state-of-the-art flexible electronics. These have sub-cellular feature sizes and actually like to interact with neurons.

Wearable electronics for brain waves

We have seen pictures of people lying in giant machines (MRI and CT-scan) that record brain activity, or a man running on a treadmill with dozens of little wires attached to his muscles. These devices making use of brain waves are usually big, expensive and can be operated only by trained people. That could be a thing of the past as we can now record the same activity using much smaller, inexpensive and easy-to-use wearable devices.

We are at a point now where we can purchase our own devices, measure our brainwaves and adjust our lives more effectively to deal with stress and improve our focus. We can attach these wearable devices to our body in order to measure physiological activity like heart rate, body temperature, respiratory rate, muscle tension, and brain waves known as biofeedback.

Imagine seeing the brain activity in real-time, while meditating or while feeling stressed at work, and being able to adapt to get better results. The activity measured using EEG can be very detailed and complex, and dependent on the area of the brain. Biofeedback headsets measure the brain waves using EEG, and these are small bands that sit easily on our head and measure activity through sensors. Some of such headsets available in the market are discussed here.

The brain sensing headband. It is an EEG headband that connects to the devices through Bluetooth to track the brainwaves. Its excellent iPhone/Android app has also been updated.

Thync calm and energy wearable. It is the first wearable device that actively elevates mood and lowers stress, and claims to boost energy levels without sugar, caffeine or other stimulants. It is worn on forehead and back of the neck. It uses electromagnetic impulses that stimulate cranial nerves on the face and back of the head to improve mood.

NeuroSky MindWave headset. It is the cheapest option that comes with



Versus headset for measuring brain waves
(Credit: <https://getversus.com>)

a big library of apps and programs to run with.

Versus headset. This is the most advanced of the wearable headsets currently available for personal use. It has the best software for brain training, and it is the most advanced biofeedback headset for measuring brain waves.

Electronics of brain waves outside the body

Many people question the existence of brain waves and say that electrical impulses in the brain don't ever get outside the body. Electrical impulses are well known to travel inside the brain and to the rest of the nervous system. But brain waves not only play an important role inside the human body, their energy can be used to make wonders outside the body as well for controlling electronic devices/displays, telekinesis, hypnotism, etc.

The energy of brain waves (as wave function) in the open will also suffer different energy losses like attenuation, absorption, dispersion and scattering, and so energy of the waves will decrease fast with distance and time. However, with strong concentration/meditation to generate brain waves of required energy level, one can perform the activities mentioned below.

Telekinesis. It is one of the bases of many super powers in human beings, which is often used for controlling/

manipulating of matter. Telekinesis is a general term for any ability that involves using the mind to influence/manipulate/move matter or objects. Further, it may evolve to the point where it can control anything at a subatomic particle to universal level.

Telepathy. The electrical nature of the brain allows sending and receiving of brain waves as electrical pulses.

These can be delivered in a non-invasive way using a technique called transcranial magnetic stimulation (TMS). Using a combination of technologies like EEG, the Internet and TMS, researchers are able to transmit a thought over a very large distance, such as from India to France.

Brain-computer interface. Recent advances in brain-computer interface are turning the science fantasy of transmitting thoughts directly from one brain to another into reality. In this field, brain-to-brain interfaces allow direct transmission of brain activity in real time by coupling the brains of two individuals. Users want to control electronic devices through brain waves without use of even hands, which brain-computer interface is now enabling.

Hypnosis. It is based on the use of brain waves for guided relaxation, intense concentration and focused attention, to achieve a heightened state of awareness that is sometimes called a spell. During hypnotic spell, a person's attention is so focused that anything going around the person is temporarily blocked out or ignored.

Working of our brain cells, or neurons, using electrical signals to communicate and coordinate for higher brain function as well as brain influence outside the brain itself are the biggest questions in all of science. Recently, researchers have been able to design an electronic chip capable of performing high-sensitivity intracellular recording involving thousands of connected neurons simultaneously, identifying hundreds of synaptic connections of brain. With such developments in science and technology, may be in near future, we will be able to understand more about the electronics of brain and brain waves. **EFY**